Official Fuzzy Hamiltonian for Circuit Design

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(Dated: November 6, 2026)

All sorts of deep shit.

I. INTRODUCTION

The wavefunction at each link is better thought as a 4×4 matrix Ψ. The hamiltonian is (lattice units):

$$\begin{aligned} H\Psi &= \kappa 1 \sum_{links\ l} [T_L^A(l),[T_L^A(l),\Psi]] + [T_R^A(l),[T_R^A(l),\Psi]] - \kappa 2 \sum_{links\ l} \left(\mathbb{U}_{ab}(l)\Psi \mathbb{U}_{ab}^\dagger(l) + \mathbb{U}_{ab}^\dagger(l)\Psi \mathbb{U}_{ab}(l)\right) \\ &+ \alpha \sum_{plaquettes\ p} \mathbb{U}_{ab}(l_1)\mathbb{U}_{bc}(l_2)\mathbb{U}_{cd}^\dagger(l_3)\mathbb{U}_{da}^\dagger(l_4)\Psi, \end{aligned} \tag{1}$$

where

$$\begin{aligned} T_R^A &= \frac{1}{2}\begin{pmatrix} \sigma^A & 0 \\ 0 & 0\end{pmatrix}, \qquad T_L^A = \frac{1}{2}\begin{pmatrix} 0 & 0 \\ 0 & \sigma^A\end{pmatrix}, \\ \mathbb{U}_{ab} &= \delta_{ab}\Gamma_0 - i\sigma_{ab}^A\Gamma_A, \qquad \Gamma_0 = \frac{1}{2}\begin{pmatrix} 0 & \mathbb{1} \\ \mathbb{1} & 0\end{pmatrix}, \quad \Gamma_A = \frac{1}{2}\begin{pmatrix} 0 & i\sigma^A \\ -i\sigma^A & 0\end{pmatrix}, \end{aligned} \tag{2}$$

$a,b,c,d = 1,2$ and $A = 1,2,3$. Gauss law:

$$[T_L^A(l_1),\Psi] + [T_L^A(l_2),\Psi] + [T_R^A(l_3),\Psi] + [T_R^A(l_4),\Psi] = 0, \tag{3}$$

where l_i, l_2 re the incoming links and l_3, l_4 are the outgoing links at that site.

We need to design circuit for

$$U_E = e^{-i\kappa 1[T_L^A(l),[T_L^A(l),\ldots]]+[T_R^A(l),[T_R^A(l),\ldots]]}, \quad U_K = e^{-i\kappa 2\left(\mathbb{U}_{ab}(l)\ldots\mathbb{U}_{ab}^\dagger(l)+\mathbb{U}_{ab}^\dagger(l)\ldots\mathbb{U}_{ab}(l)\right)}, \quad U_B = e^{-i\alpha \mathbb{U}_{ab}(l_1)\mathbb{U}_{bc}(l_2)\mathbb{U}_{cd}^\dagger(l_3)\mathbb{U}_{da}^\dagger(l_4)\ldots}. \tag{4}$$

U_E and U_K are one link operators and should be very simple. U_B is a 4-link (plaquette) operator. If we assume the incoming state satisfies Gauss law, the final state will also satisfy it as each term is gauge invariant. A simple way of encoding the Hilbert space of this theory is to use 4-qubits per link as $2^4 = 16$ dimensions of the one-site Hilbert space. Thsi is not the most economic way because most of these state do not satisfy Gauss law, but it's a place to start. I'll think of something cleverer.

ACKNOWLEDGMENTS

P. B. is supported by the U.S. Department of Energy under Contract No. DE-FG02-93ER-40762.

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